SEMI NEX AI

AI-Powered Semiconductor Quality Intelligence & Decision Support

DEFECT INSPECTION REPORT

CONFIDENTIAL & AUDIT READY

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Inspection ID:	INSP-20261023-CCCF67	**Model Version:**	v1.2.0-spatial-gbc
Timestamp:	2026-10-23T05:51:00.758056	**Operating Mode:**	**REAL MODEL INFERENCE**
Lot Identifier:	lot42000	**Wafer Index:**	2
Input Source:	WM-811K Dataset Record	**Original Dataset Label:**	Center

1. Classification & Triage Recommendation

Predicted Defect Class:	**Center**	**Classification Confidence:**	**100.0%**
Triage Recommendation:	**Review Recommended**	**Review Status:**	**PENDING**
Decision Justification:	Moderate pattern risk (35.7/100) or borderline classification confidence (100.0%). Secondary inspection verification recommended.	**Human Notes:**	No notes submitted.

2. Spatial Analysis & Wafer Topography Metrics

Total Wafer Dies:	912	**Failing Die Count:**	105
Wafer Yield Rate:	**88.49%**	**Failure Proportion:**	**11.51%**

3. Possible Root Cause / Hypothesis & Engineering Guidance

Working Hypothesis: Center-Concentrated Process Anomaly

Suggested Verification Directions:

- Conduct optical inspection of wafer bevel and perimeter clamp marks.
- Review recent electrostatic chuck (ESC) temperature stability and helium leak check logs.
- Verify chamber preventive maintenance schedule and focus ring RF lifetime hours.

LEGAL & AUDIT DISCLAIMER — NOT A SUBSTITUTE FOR PROFESSIONAL FAB INSPECTION:
SEMI NEX AI is an AI-assisted semiconductor intelligence and decision support platform. Predictions and risk scores are derived from topological wafer map die matrices and machine learning models. Wafer map patterns alone do not establish physical causes, electrical performance, or contamination provenance. Autonomous lot disposition is not supported; critical decisions require qualified quality engineering verification.